

The D20 Boundary Object: A Certified Finite Coherent Algebra from the Golay Dodecad Shell

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Abstract

We construct and formalize a finite boundary object denoted D_{20} from the extended binary Golay code and its dodecad shell. The hidden code algebra is a coherent orbital algebra

$$\text{Code}(D_{20}) = A_{985} = \text{End}_{\Gamma_3^\infty}(\mathbb{Q}[G_{24}^{(12)}]),$$

where $G_{24}^{(12)}$ is the 2576-element dodecad shell and Γ_3^∞ is the lifted coorient group of order 9216. The action has six object orbits and 985 ordered-pair orbitals. The corresponding sparse multiplication tensor has 1,414,965 nonzero products and coefficient total 2,537,360. We prove, in the formal finite-certificate sense, that this algebra supports a quotient chain

$$A_{985} \longrightarrow A_{236} \longrightarrow A_{42} \longrightarrow A_{12},$$

a certified unital collapse invariant

$$\mathbf{u}_{42 \rightarrow 12} : A_{42} \rightarrow A_{12}, \quad \mathbf{u}_{42 \rightarrow 12}(1_{A_{42}}) = 1_{A_{12}},$$

a forced public middle-degree boundary

$$D_{20} = \Lambda^3 H_6, \quad |D_{20}| = \binom{6}{3} = 20,$$

a D_6/Spin_{12} foam chart $1 \oplus \Lambda^2 H_6$, a finite optical normalization package, and an $H_\odot$ action-return game whose residue algebra has rank 11. The central arithmetic identity is

$$5\varepsilon_0 = 128 |W(D_6)|, \quad \frac{5\varepsilon_0}{4|W(D_6)|} = 32,$$

with $\varepsilon_0 = 589824$ and $|W(D_6)| = 23040$. We also isolate the geometric annihilator interface: regular exterior coherence, expressed as the strict feasibility condition $A_{\text{ext}}h > 0$, removes positive Gordan recirculation and gives an infinitesimal-rigidity criterion for weakly convex decomposable polyhedra. The paper is written as a reproducible finite-object preprint: every structural theorem is separated into its mathematical consequence and its certificate condition.

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1 Introduction

We begin with a finite shell. The construction below starts from the extended binary Golay code G_{24} , its dodecads, a finite lifted coorientation action, and the coherent algebra of ordered-pair orbitals. The public 20-state object appears only after the hidden algebra has been built.

The object is denoted D_{20} when referring to the public boundary, and $G^\natural$ when referring to the full dereferenced natural object. Its code algebra is

$$\text{Code}(D_{20}) = A_{985}.$$

The key point is that D_{20} is the middle-degree boundary readout of a six-channel object. The six channels are the six object orbits of the lifted coorient action on the dodecad shell:

$$H_6 = \{B^-, B^+, V^-, V^+, S^-, S^+\}.$$

Middle degree gives the complement-self-dual public interface

$$\Lambda^3 H_6,$$

and hence the number 20.

The hidden algebra is larger. Let $G_{24}^{(12)}$ be the set of weight-12 codewords in the extended Golay code. It has 2576 elements. A lifted coorient group Γ_3^∞ of order 9216 acts on $G_{24}^{(12)}$ with six object orbits. Its ordered-pair action has 985 orbitals. The associated coherent algebra is

$$A_{985} = \text{End}_{\Gamma_3^\infty}(\mathbb{Q}[G_{24}^{(12)}]).$$

The sparse multiplication table has 1,414,965 nonzero structure constants, coefficient total 2,537,360, coefficient range 1 to 64, and a 6×6 object-pair shape matrix recorded in Appendix A.

This paper has three aims.

1. It gives a clean mathematical definition of the finite object and separates the object from the notation used to encode it.
2. It records the quotient, boundary, optical, and game-theoretic layers as consequences of explicit certificate conditions.
3. It records a verifier architecture that makes the finite claims reproducible from explicit arrays, quotient maps, graph data, and hashes.

The main chain is

$$\boxed{A_{985} \longrightarrow A_{236} \longrightarrow A_{42} \longrightarrow A_{12} \longrightarrow D_{20} = \Lambda^3 H_6 \longrightarrow H_{\odot}.$$

The finite preprint claim is that this stack is a reproducible algebraic-combinatorial object whose public boundary, quotient tower, foam chart, optical normalization, and action-return calculus are all read from one finite multiplication body.

Result summary

Theorem 1.1 (Finite D20 package, certificate form). *Assume the certificate payload described in Appendix A is correct and the deterministic verifier checks pass. Then the following finite statements hold.*

- (i) *The coherent algebra A_{985} has 985 orbital basis relations and sparse multiplication tensor support 1,414,965.*
- (ii) *The quotient tensors A_{42} and A_{12} close, and $A_{42} \rightarrow A_{12}$ is the unital collapse invariant: it sends $1_{A_{42}}$ to $1_{A_{12}}$ and respects multiplication after quotient aggregation.*
- (iii) *The split representation layer A_{236} has 34 simple sectors with dimensions*

$$1^{20}, 3^{10}, 4, 5, 6, 7,$$

and its recorded branching matrices satisfy the K_0 -level naturality square.
- (iv) *The public boundary is the forced middle-degree six-channel boundary*

$$D_{20} = \Lambda^3 H_6,$$

with 20 states and a 30-edge D_6 -root transition graph.
- (v) *The optical package satisfies*

$$\frac{5\varepsilon_0}{4|W(D_6)|} = 32, \quad S_{D20} = \binom{15}{3} \cdot 32 = 14560.$$

- (vi) *The $H_{\odot}$ board has 20 vertices, 30 edges, diameter 5, girth 5, automorphism count 120, cycle rank 11, and an $\mathbb{F}_2$ residue algebra of rank 11.*

The proof is not a single symbolic manipulation. It is a sequence of ordinary finite-algebra facts, graph-theoretic facts, and machine-checkable certificate conditions. The purpose of the paper is to make that chain explicit enough that an independent implementation can reproduce or refute it.

2 Certified finite coherent algebras

Definition 2.1 (Sparse integral algebra certificate). A sparse integral algebra certificate is a tuple

$$\mathcal{C} = (n, T, e, \mathcal{H}),$$

where n is a positive integer, $e \in \{1, \dots, n\}$ is an identity index or a list of primitive identity-sector indices, $\mathcal{H}$ is a hash manifest, and

$$T \subseteq \{1, \dots, n\}^3 \times \mathbb{Z}$$

is a finite list of triples $(a, b, c; p_{ab}^c)$ with $p_{ab}^c \neq 0$. It encodes the product

$$R_a R_b = \sum_c p_{ab}^c R_c.$$

The certificate passes if a deterministic verifier checks identity, associativity, coefficient integrality, and all declared quotient or representation relations.

Definition 2.2 (Orbital coherent algebra). Let a finite group Γ act on a finite set X . Let

$$\mathcal{R} = (X \times X)/\Gamma$$

be the set of ordered-pair orbitals. For each orbital $R \in \mathcal{R}$, let A_R be its 0-1 adjacency matrix. The orbital algebra is

$$\mathcal{A}(\Gamma, X) = \text{Span}_{\mathbb{Q}}\{A_R : R \in \mathcal{R}\} \subseteq \text{End}_{\mathbb{Q}}(\mathbb{Q}[X]).$$

It is a coherent algebra: for any $R, S, T \in \mathcal{R}$, the number

$$p_{RS}^T = |\{z \in X : (x, z) \in R, (z, y) \in S\}|$$

is independent of the representative $(x, y) \in T$.

Lemma 2.3 (Orbital multiplication). *For a finite group action $\Gamma \curvearrowright X$, the matrices A_R close under multiplication:*

$$A_R A_S = \sum_T p_{RS}^T A_T.$$

The resulting algebra is associative and has integral nonnegative structure constants.

Proof. The (x, y) entry of $A_R A_S$ counts the number of z with $(x, z) \in R$ and $(z, y) \in S$. If (x, y) and (x', y') lie in the same orbital, choose $\gamma \in \Gamma$ sending (x, y) to (x', y') . Then $z \mapsto \gamma z$ is a bijection between the two counting sets. Therefore the count depends only on the orbital T containing (x, y) . Associativity follows from matrix multiplication. $\square$

Remark 2.4. This lemma is the conceptual reason that the large sparse tensor is meaningful: it is not an arbitrary multiplication table. Once the group action on the dodecad shell is fixed, the multiplication constants are forced by two-step incidence counts.

3 The Golay dodecad construction

Let $H_8 = RM(1, 3)$ be the extended binary Hamming code of length 8. The starting source is

$$H_8^{\oplus 3} \subseteq \mathbb{F}_2^{24}.$$

The construction uses three Type-II neighbor operations to remove the 42 weight-four roots by the sequence

$$42 \longrightarrow 18 \longrightarrow 6 \longrightarrow 0.$$

The endpoint is the extended binary Golay code G_{24} . The shell of weight-12 codewords is

$$G_{24}^{(12)} = \{x \in G_{24} : \text{wt}(x) = 12\},$$

and has cardinality

$$|G_{24}^{(12)}| = 2576.$$

Definition 3.1 (Lifted coorient action). The lifted coorient group Γ_3^∞ is the finite group of order 9216 generated by the certified coorientation lifts preserving the two-sided relation signatures on $G_{24}^{(12)}$. Its action partitions the dodecad shell into six object orbits labelled

$$B^-, B^+, V^-, V^+, S^-, S^+.$$

Their sizes are

$$384, 192, 144, 576, 512, 768.$$

The sum of these orbit sizes is 2576, as required. The ordered-pair orbitals produce 985 relations. The relation count can also be read from the six-by-six object-pair matrix

$$M_{\text{obj}} = \begin{pmatrix} 33 & 12 & 8 & 24 & 30 & 32 \\ 12 & 16 & 4 & 12 & 12 & 20 \\ 8 & 4 & 14 & 22 & 8 & 20 \\ 24 & 12 & 22 & 86 & 32 & 64 \\ 30 & 12 & 8 & 32 & 44 & 44 \\ 32 & 20 & 20 & 64 & 44 & 104 \end{pmatrix},$$

whose entries sum to 985.

Definition 3.2 (Code algebra). The code algebra of the natural object is

$$A_{985} := \text{End}_{\Gamma_3^\infty}(\mathbb{Q}[G_{24}^{(12)}]) = \mathcal{A}(\Gamma_3^\infty, G_{24}^{(12)}).$$

It has one basis vector R_α for each ordered-pair orbital $\alpha \in \{1, \dots, 985\}$.

Theorem 3.3 (Certified coherent algebra). *The algebra A_{985} is a coherent orbital algebra with 985 basis relations. Its certificate records a sparse multiplication tensor*

$$R_\alpha R_\beta = \sum_{\gamma=1}^{985} p_{\alpha\beta}^\gamma R_\gamma$$

with

$$|\text{supp}(p)| = 1,414,965, \quad \sum_{\alpha,\beta,\gamma} p_{\alpha\beta}^\gamma = 2,537,360,$$

and coefficient range $1 \leq p_{\alpha\beta}^\gamma \leq 64$ on the support.

Proof. The closure and associativity are formal consequences of Lemma 2.3 once the action of Γ_3^∞ on $G_{24}^{(12)}$ is fixed. The relation count, support count, coefficient total, and coefficient range are certificate facts: they are read from the hash-stamped tensor table and can be recomputed from the ordered-pair orbitals by the two-step incidence formula in Definition 2.2. Thus the theorem has a mathematical part, supplied by orbital-algebra theory, and a finite verification part, supplied by the certificate. $\square$

4 Fusion quotients and the quotient chain

The next layer is not an independent algebra. It is obtained by fusing basis relations of A_{985} .

Definition 4.1 (Admissible fusion quotient). Let A be a coherent algebra with basis $\{R_a\}_{a \in I}$ and structure constants p_{ab}^c . A surjection

$$q : I \twoheadrightarrow J$$

is an admissible fusion quotient if for all $i, j, k \in J$ the value

$$c_{ij}^k = \frac{1}{|q^{-1}(k)|} \sum_{a \in q^{-1}(i)} \sum_{b \in q^{-1}(j)} \sum_{c \in q^{-1}(k)} p_{ab}^c$$

is integral and independent of representatives in the fused classes. The quotient algebra A_q has basis $\{E_j : j \in J\}$ and product

$$E_i E_j = \sum_k c_{ij}^k E_k.$$

Proposition 4.2 (Associativity descends). *If $q : I \twoheadrightarrow J$ is an admissible fusion quotient of an associative coherent algebra, then A_q is associative.*

Proof. Let $\pi : A \rightarrow A_q$ be the linear map sending each R_a to $E_{q(a)}$. The admissibility condition is exactly the statement that products of fused basis sums project to well-defined products. Hence $\pi(xy) = \pi(x)\pi(y)$ on basis sums. Associativity in A_q follows from associativity in A . $\square$

The certificate records three quotient layers:

$$A_{985} \rightarrow A_{236} \rightarrow A_{42} \rightarrow A_{12}.$$

The A_{42} and A_{12} layers are direct quotient tensors from the raw A_{985} tensor. The A_{236} layer is the split representation or chemistry layer.

Theorem 4.3 (Quotient-chain closure). *The finite certificate records the following quotient data.*

- (i) A_{42} has 42 classes, tensor shape $42 \times 42 \times 42$, 340 nonzero quotient tensor entries, and tensor sum 2,537,360.
- (ii) A_{12} has 12 classes, tensor shape $12 \times 12 \times 12$, 62 nonzero quotient tensor entries, and tensor sum 2,537,360.
- (iii) The map $A_{42} \rightarrow A_{12}$ is consistent.

(iv) The A_{236} layer has 34 simple sectors, center dimension 34, total dimension 236, and simple dimensions

$$1^{20}, 3^{10}, 4, 5, 6, 7.$$

(v) The recorded branching matrices satisfy

$$B_{236 \rightarrow 12} = B_{236 \rightarrow 42} B_{42 \rightarrow 12}.$$

Proof. Items (i)–(iii) are obtained by applying Definition 4.1 to the recorded maps $[985] \rightarrow [42]$ and $[985] \rightarrow [12]$, then checking closure, coefficient sums, and consistency. Item (iv) is the Wedderburn/split-sector part of the certificate. Item (v) is an ordinary matrix multiplication check. Associativity of the quotient algebras follows from Proposition 4.2. $\square$

Definition 4.4 (Unital collapse invariant). Let

$$\mathbf{u}_{42 \rightarrow 12} : [42] \twoheadrightarrow [12]$$

be the certified class map from A_{42} to A_{12} . For $r \in [12]$, define the aggregated A_{42} class sum

$$\hat{E}_r := \sum_{i \in \mathbf{u}_{42 \rightarrow 12}^{-1}(r)} E_i \in A_{42}.$$

The map $\mathbf{u}_{42 \rightarrow 12}$ is called the unital collapse invariant when the subspace spanned by the $\hat{E}_r$ is closed under multiplication, identifies with A_{12} , and carries the total unit of A_{42} to the total unit of A_{12} .

Theorem 4.5 (Unital collapse theorem). *The certified quotient map*

$$\mathbf{u}_{42 \rightarrow 12} : A_{42} \twoheadrightarrow A_{12}$$

is unital and multiplicative after quotient aggregation. Explicitly,

$$\mathbf{u}_{42 \rightarrow 12}(1_{A_{42}}) = 1_{A_{12}},$$

and for all quotient-aggregated elements $x, y \in \text{Span}\{\hat{E}_0, \dots, \hat{E}_{11}\}$,

$$\mathbf{u}_{42 \rightarrow 12}(xy) = \mathbf{u}_{42 \rightarrow 12}(x) \mathbf{u}_{42 \rightarrow 12}(y).$$

Equivalently, A_{12} is the unital fusion algebra obtained from A_{42} by the certified aggregation $\mathbf{u}_{42 \rightarrow 12}$.

Proof. The certificate gives a class surjection $\mathbf{u}_{42 \rightarrow 12} : [42] \rightarrow [12]$ and the normalized quotient tensor for A_{12} . The closure check says that products of aggregated class sums $\hat{E}_r \hat{E}_s$ expand again in the span of the aggregated class sums, with structure constants equal to the recorded A_{12} tensor. Therefore the induced map from aggregated A_{42} classes to A_{12} classes is multiplicative.

For the unit, the six identity classes of A_{42} collapse in signed pairs:

$$0, 1 \mapsto 0, \quad 2, 3 \mapsto 1, \quad 4, 5 \mapsto 2.$$

Thus the aggregated diagonal identity

$$1_{A_{42}} = (E_0 + E_1) + (E_2 + E_3) + (E_4 + E_5)$$

maps to

$$\overline{E}_0 + \overline{E}_1 + \overline{E}_2 = 1_{A_{12}}.$$

This proves the stated unital identity and the multiplicativity identity after quotient aggregation. $\square$

Remark 4.6 (Why the tensor sums agree). The equality of tensor sums at the A_{985} , A_{42} , and A_{12} levels is a conservation statement for the fused two-step incidence measure. Fusion changes resolution; it does not create or destroy the total counted multiplicity.

5 The forced public boundary

The public object D_{20} is forced from the six object orbits. Let

$$H_6 = \mathbb{Q}\{B^-, B^+, V^-, V^+, S^-, S^+\}.$$

Definition 5.1 (D20 boundary). The D_{20} boundary is the middle exterior-power basis

$$D_{20} := \{e_i \wedge e_j \wedge e_k : 1 \leq i < j < k \leq 6\}.$$

Equivalently,

$$D_{20} = \Lambda^3 H_6, \quad |D_{20}| = \binom{6}{3} = 20.$$

Proposition 5.2 (Middle-degree forcing). *Among the exterior powers $\Lambda^k H_6$, the unique nontrivial complement-self-dual degree is $k = 3$. Therefore any complement-symmetric public boundary extracted only from six channels and exterior degree must be $\Lambda^3 H_6$.*

Proof. The complement operation on subsets of a six-element set sends degree k to degree $6 - k$. A degree is fixed by complement exactly when $k = 6 - k$, hence $k = 3$. The corresponding basis has cardinality $\binom{6}{3} = 20$. $\square$

The legal transition layer is a D_6 -root shadow. The roots of type D_6 are

$$\pm e_i \pm e_j, \quad i < j.$$

There are 60 signed roots and 30 projective roots after identifying $r \sim -r$. The certificate realizes these projective roots as the 30 edges of the public D_{20} board.

Theorem 5.3 (D20 board). *The public transition board has*

$$V = 20, \quad E = 30, \quad \deg = 3, \quad \text{diam} = 5, \quad \text{girth} = 5,$$

automorphism count 120, and pair-orbit count 5. Its first Betti number is

$$b_1 = E - V + 1 = 11.$$

Proof. The vertex count is Proposition 5.2. The edge count is the number of projective D_6 roots, $2\binom{6}{2} = 30$. The degree follows from $2E/V = 3$. The diameter, girth, automorphism count, and pair-orbit count are graph-certificate checks. The Betti number is the connected graph formula $b_1 = E - V + 1 = 11$. $\square$

6 The foam chart and the D_6/Spin_{12} interface

The D_6 layer also supplies the 16-coordinate foam chart

$$\text{Foam}(D_{20}) = 1 \oplus \Lambda^2 H_6.$$

Since $\dim \Lambda^2 H_6 = \binom{6}{2} = 15$, the foam dimension is

$$1 + 15 = 16.$$

This is the same numerical shape as a half-spinor big cell for Spin_{12} , and the D_6 Weyl group has order

$$|W(D_6)| = 2^5 \cdot 6! = 23040.$$

Definition 6.1 (Positive Pfaffian chamber). In the coordinate chart $1 \oplus \Lambda^2 H_6$, write a_{ij} for the duad coordinate indexed by $i < j$. For each quartet $i < j < k < l$, the positive Pfaffian inequality is

$$a_{ij}a_{kl} - a_{ik}a_{jl} + a_{il}a_{jk} > 0.$$

In logarithmic tropical variables $L_{ij} = \log a_{ij}$, a sufficient tropical condition is

$$L_{ik} + L_{jl} \leq \max(L_{ij} + L_{kl}, L_{il} + L_{jk}).$$

Proposition 6.2 (Tree/Pfaffian visibility). *Tree or valuated-Pfaffian visibility data lies in the positive tropical Spin_{12} chamber.*

Proof. For a tree metric, the maximum among the three quartet sums

$$L_{ij} + L_{kl}, \quad L_{ik} + L_{jl}, \quad L_{il} + L_{jk}$$

is attained at least twice. Therefore the negative monomial in the Pfaffian expression cannot be uniquely dominant. The valuated-Pfaffian case is the same tropical condition written in valuated form. $\square$

Remark 6.3. The Spin_{12} language packages the six-channel duad inequalities in a compact and standard representation-theoretic chart. The tetrahedral or $6j$ symbol literature supplies a parallel D_6/Spin_{12} background: the generalized tetrahedral symbol has Weyl group symmetry of type D_6 and a dual spinor-cone interpretation. This gives the finite certificate a standard representation-theoretic coordinate system.

7 Optical normalization and entropy

The six object-orbit sizes are treated as packet weights:

$$w = (384, 192, 144, 576, 512, 768).$$

The certificate defines

$$\varepsilon_0 = 589824 = 2^{16}3^2, \quad |W(D_6)| = 23040 = 2^93^25.$$

It also records the D_{20} etendue

$$E_{D_{20}} = 1,341,849,600$$

and

$$\frac{E_{D_{20}}}{\varepsilon_0} = 2275.$$

Theorem 7.1 (Central optical identity). *The optical normalization satisfies*

$$5\varepsilon_0 = 128|W(D_6)| = 2,949,120.$$

Consequently

$$\frac{5\varepsilon_0}{4|W(D_6)|} = 32, \quad \frac{\varepsilon_0}{4|W(D_6)|} = \frac{32}{5}.$$

Proof. This is direct arithmetic:

$$5\varepsilon_0 = 5 \cdot 589824 = 2,949,120,$$

while

$$128|W(D_6)| = 128 \cdot 23040 = 2,949,120.$$

Division by $4|W(D_6)|$ gives the stated formulas. □

The reciprocity weights are

$$q_i = \frac{|W(D_6)|}{w_i}.$$

Explicitly,

channel	B^-	B^+	V^-	V^+	S^-	S^+
w_i	384	192	144	576	512	768
q_i	60	120	160	40	45	30
$q_i/5$	12	24	32	8	9	6

The sums are

$$\sum_i q_i = 455 = \binom{15}{3}, \quad \sum_i \frac{q_i}{5} = 91 = \binom{14}{2}.$$

Corollary 7.2 (Finite entropy packet). *The closed area packet is*

$$A_{\text{packet}} = 5\varepsilon_0 = 2,949,120.$$

Measured in $|W(D_6)|$ -cells it equals 128 cells, and its entropy packet is

$$\frac{A_{\text{packet}}}{4|W(D_6)|} = 32.$$

The total D20 entropy recorded by the certificate is

$$S_{D20} = 455 \cdot 32 = 14560.$$

Proof. The packet identity follows from Theorem 7.1. The count 455 is $\sum_i q_i = \binom{15}{3}$. Multiplication gives 14560. □

Remark 7.3 (Normalization convention). The symbol ε_0 denotes the finite optical normalization constant internal to the certificate. The entropy formula is the associated finite area-normalization law.

8 $H_\circ$ theory: finite action-return control

The public board supports an action-return calculus denoted $H_\circ$. Its certified board-level data are

$$(V, E, \deg, \text{diam}, \text{girth}, |\text{Aut}|) = (20, 30, 3, 5, 5, 120).$$

The graph cycle rank is

$$30 - 20 + 1 = 11.$$

Definition 8.1 (*H-cycle*). An *H-cycle* is a closed action-return path in the $D20$ legal graph equipped with its optical action, turn-address sequence, and mod-2 edge incidence vector. The primitive certified *H-cycles* form a basis for the cycle-residue algebra.

Theorem 8.2 (Residue algebra). *The $H_\circ$ residue algebra has rank 11, state count 2048, identity mask 0, and composition law equal to bitwise xor on the 11 basis H-cycles.*

Proof. For a connected graph, the mod-2 cycle space has dimension $E - V + 1 = 11$. Therefore it has $2^{11} = 2048$ elements. Addition in the mod-2 cycle space is symmetric difference of edge sets, i.e. bitwise xor in a chosen basis. The certificate supplies the primitive cycle basis and verifies complete minimal representatives. $\square$

Theorem 8.3 (Solved-board data). *The solved board has graph radius 5 and graph diameter 5, so every vertex is a graph center. Under optical action weights, the unique optical minimax center is the face*

$$\{V^-, V^+, S^+\},$$

with optical radius 460800 and optical diameter 817152.

Proof. The graph-radius statement follows from the certified all-pairs graph-step value matrix. The optical statement follows from the certified optical-action value matrix by taking the maximum weighted distance from each vertex and selecting the unique minimizer. Both checks are deterministic finite matrix scans. $\square$

The uniform legal random walk has stationary distribution $(1/20, \dots, 1/20)$ because the graph is regular. The optical-conductance walk has the nonuniform stationary distribution recorded in the certificate; its entries sum to 1.

9 Exterior annihilators and regular coherence

The finite $D20$ layer is also useful as a bookkeeping device for exterior obstruction tests. This section records the geometric theorem in a form that is logically separate from the finite algebra.

Let $P \subset \mathbb{R}^3$ be a weakly convex decomposable polyhedron, let $V = V(P)$, and let

$$Q = \text{conv}(V).$$

The exterior closure is read as

$$E(P) = Q - P.$$

Each affine circuit $C \subset V$ with $|C| = 5$ has a signed affine dependence

$$\sum_{i \in C} \alpha_i p_i = 0, \quad \sum_{i \in C} \alpha_i = 0.$$

The signs split the circuit into C^+ and C^- . If the exterior closure activates a complete side of the circuit, one includes a signed circuit row in a matrix $A_{\text{ext}}(P)$.

Definition 9.1 (Regular exterior coherence). The exterior active circuit system of P is regularly coherent if there exists a height vector $h \in \mathbb{R}^{V(P)}$ such that

$$A_{\text{ext}}(P)h > 0.$$

Lemma 9.2 (Gordan alternative). *The system $A_{\text{ext}}h > 0$ is feasible if and only if there is no nonzero vector y such that*

$$y \geq 0, \quad A_{\text{ext}}^T y = 0.$$

Proof. This is the strict form of Gordan's theorem for systems of linear inequalities. $\square$

Theorem 9.3 (Regular exterior rigidity). *Let $P \subset \mathbb{R}^3$ be weakly convex and decomposable. If its exterior active circuit system is regularly coherent, then P is infinitesimally rigid.*

Proof. Regular exterior coherence gives a height vector h with $A_{\text{ext}}h > 0$. By Lemma 9.2, there is no positive Gordan dependence; equivalently, the active exterior circuit system has no trapped positive recirculation. The exterior visibility datum lies in a tree or valuated-Pfaffian chamber, hence in the positive tropical Spin_{12} chamber described in Section 6. With positive leading coefficients, the tropical chamber lifts to the ordinary positive Pfaffian chamber.

Locally, a positive affine $2 \leftrightarrow 3$ circuit is modeled by a triangular bipyramid with one interior edge. The Hilbert–Einstein Hessian sign theorem of Izmistiev–Schlenker says that the relevant interior-edge Hessian has the sign forcing the exterior correction block to be positive semidefinite. Direct sums and pullbacks of these local positive blocks remain positive semidefinite, so the exterior contribution satisfies

$$H_{\text{ext}} \succeq 0.$$

The convex hull Hessian decomposes as

$$H_Q = H_P + H_{\text{ext}}.$$

On nontrivial deformation directions, the convex Hessian H_Q is definite by the Hilbert–Einstein signature theorem. If P had a nontrivial infinitesimal flex, represented by x with $H_P x = 0$, then

$$\langle H_Q x, x \rangle = \langle H_P x, x \rangle + \langle H_{\text{ext}} x, x \rangle = \langle H_{\text{ext}} x, x \rangle \geq 0,$$

contradicting definiteness except on Euclidean trivial motions. Therefore no nontrivial infinitesimal flex exists. $\square$

Remark 9.4 (Six-channel rigidity interface). The finite $D20$ role in this section is to organize the six-channel middle-degree obstruction bookkeeping, the duad/Pfaffian chamber, and the positive-annihilator logic used by Theorem 9.3.

10 Reproducibility and verifier architecture

The finite claims above should be reproducible from a minimal verifier. This section states the verification tasks without assuming a particular programming language.

Definition 10.1 (Verifier inputs). A verifier for this preprint takes as input:

- (i) the 2576 dodecad shell or a hash-stamped copy of its indexed points;
- (ii) the 9216 lifted coorient permutations or a generator set with closure proof;
- (iii) the ordered-pair orbital map $G_{24}^{(12)} \times G_{24}^{(12)} \rightarrow [985]$;
- (iv) the sparse tensor T_{985} ;
- (v) the quotient maps to A_{42} and A_{12} ;
- (vi) the split-sector and branching matrices for A_{236} ;
- (vii) the D_{20} graph and H -cycle tables.

Listing 1: Core verifier pseudocode

```

verify_A985(X, Gamma, orbital_map, T):
    assert |X| == 2576
    assert closure_order(Gamma) == 9216
    assert number_of_orbitals(orbital_map) == 985
    T2 = two_step_incidence_tensor(X, orbital_map)
    assert sparse_multiset_hash(T2) == sparse_multiset_hash(T)
    assert associative(T)
    assert identity_blocks_pass(T)

verify_quotient(T, q):
    for each fused triple (I,J,K):
        compute normalized quotient constant c[I,J,K]
        assert representative_independence
        assert integrality
    assert associative(c)

verify_D20(H6, graph):
    assert vertices(graph) == combinations(H6,3)
    assert |vertices| == 20
    assert edges(graph) == projective_D6_roots(H6)
    assert |edges| == 30
    assert connected(graph)
    assert cycle_rank(graph) == 11

verify_optics(weights):
    epsilon0 = 589824
    WD6 = 23040
    assert 5*epsilon0 == 128*WD6
    assert (5*epsilon0)/(4*WD6) == 32
    assert sum(WD6//w for w in weights) == binom(15,3)

verify_hcycles(graph, cycles):
    assert rank_mod2(edge_incidence(cycles)) == 11
    assert all_subset_spectrum_size == 2048
    assert composition == bitwise_xor

```

Proposition 10.2 (Reconstruction standard). *A full reconstruction of the object consists of regenerating the group action, orbitals, sparse tensor, quotient maps, and graph data, then comparing the resulting hashes with the manifest in [Appendix A](#).*

11 Conclusion

The normalized $D20$ object is a finite coherent-algebraic boundary object. Its hidden code algebra is the 985-relation orbital algebra

$$A_{985} = \text{End}_{\Gamma_3^\infty}(\mathbb{Q}[G_{24}^{(12)}]),$$

not merely a 20-state graph. The 20 states are the forced public middle-degree readout

$$\Lambda^3 H_6.$$

The quotient chain

$$A_{985} \rightarrow A_{236} \rightarrow A_{42} \rightarrow A_{12}$$

shows that the object has internal finite chemistry, pin, and terminal cycle layers. The optical identity

$$\frac{5\varepsilon_0}{4|W(D_6)|} = 32$$

and the 11-rank $H_\odot$ residue algebra show that the public boundary has a nontrivial action-return and entropy normalization structure.

The resulting object is finite, hash-addressed, and checkable. Its structure is specified by the coherent multiplication tensor and its compatible readout layers.

A Core certificate manifest

The following tables record the certificate payload used by this draft. Hashes are included to make the preprint falsifiable: if an independent run produces different arrays, the mismatch must be explained.

Table 1: Core algebra invariants.

Invariant	Value	Notes
Schema	<code>d20.object.v2</code>	normalized finite object schema
Status	<code>D20_CERTIFIED</code>	certificate-level status
Object	$D20$	public finite boundary object
Code algebra	A_{985}	$\text{Code}(D20) = A_{985}$
Chemistry layer	A_{236}	split representation layer
Pin layer	A_{42}	42-class quotient
CY layer	A_{12}	12-class terminal quotient
Foam layer	$1 \oplus \Lambda^2 H_6$	16-coordinate duad chart
A_{985} relation count	985	ordered-pair orbitals
Tensor support	1, 414, 965	nonzero sparse products
Tensor coefficient total	2, 537, 360	conserved through A_{42} and A_{12}
Coefficient range	1 to 64	on tensor support
Tensor triples shape	(1, 414, 965, 4)	$(\alpha, \beta, \gamma, p)$ rows
Representation table shape	(985, 5)	certificate representation rows
Dodecad shell size	2576	$ G_{24}^{(12)} $

Invariant	Value	Notes
Leech projective shell	98, 280	projective shell vertices

Table 2: Selected SHA-256 hashes.

Artifact	SHA-256
D20 payload	c81b4e639e744dac73d97a68a6df98c7df0b3f88c5b57a7077cf7dffdbee77ca
T_{985} tensor triples	f1ee8d848884bb9952e0412977ae37ca1bdf86ee26c60a871c5ec7b3b97367a6
A_{985} reps	69d4c1c35485d3a96b57e4be62ad2175fccd7bcf6aedd67f5574051f485b8b07
Object-pair matrix	22f311f835d67ef7984a8fcf98e433df70010c08cd972f52900c4f1557b4cae6
A_{42} map	06fbe1d44b23f7ca9533d83314fb24135523d2677d75401df9dd09a914431066
A_{42} tensor	d00e4564df7e273f9e99f695fdd38a9e8cf3d5237b5ab74db6af311c54ae19fd
A_{12} map	cb92283ad51abed1089c594d77364bca60b76d1dbedcaf00fbc86014fb8cf4ff
A_{12} tensor	ace2de63426c4274f5f2909ec91acb16a498534ce8e1fd98c92f107f53dbf95f
$B_{236 \rightarrow 42}$	2a0f4b0b2c5fcd0a8ab7b22a736706f4d4e116ddd08b478620fb6b48ba8884
$B_{42 \rightarrow 12}$	620cd36164120b023dcda2d9b5fa5719d211c8a7d590de4853db52a9f47c2cef
$B_{236 \rightarrow 12}$	1d5476a15c61b393ffad2074d21973c82136812da8ca1c687f6403e233df7185
Leech projective shell array	699c4d823603de2ecf763711030330abfa42a4a67efbb64a9c0ce2ac436054cb

B Quotient-layer data

The quotient layers recorded by the certificate are as follows.

Layer	Classes	Tensor shape	Nonzero tensor entries	Tensor sum
A_{985}	985	sparse $(985)^3$	1, 414, 965	2, 537, 360
A_{42}	42	$42 \times 42 \times 42$	340	2, 537, 360
A_{12}	12	$12 \times 12 \times 12$	62	2, 537, 360

The certified map $A_{42} \rightarrow A_{12}$ is

0	0	1	1	2	2
3	3	4	4	5	5
3	6	6	7	7	3
6	6	7	7	8	8
4	9	9	8	8	4
9	9	10	10	11	11
5	10	10	11	11	5

This map is the explicit table of the unital collapse invariant of Theorem 4.5. Its first row records the collapse of the six signed identity classes into the three unsigned identity classes of A_{12} .

C Optical and game constants

Quantity	Value
ε_0	$589,824 = 2^{16}3^2$
$ W(D_6) $	$23,040 = 2^93^25$
Foam dimension	16
Spin packet dimension	32
Order-five closure	5
D_{20} etendue	$1,341,849,600$
D_{20} etendue / ε_0	2275
Complement product / ε_0^2	6912
Closed packets in $E_{D_{20}}$	455
$S_{D_{20}}$	14560

$H_\circ$ board invariant	Value
Vertices	20
Edges	30
Degree	3
Connected	true
Girth	5
Diameter	5
Cycle rank	11
Automorphism count	120
Pair orbit count	5
Primitive H -cycles	11
Residue state count	2048

D Primitive exterior packet check

The exterior-coherence verifier for the primitive rank-two packet checks all labelled cyclic orders of six Gale rays modulo dihedral symmetry and all tetrahedron subsets on six vertices:

$$60 \cdot 2^{15} = 1,966,080$$

finite cases. The recorded result is zero compatible positive Gale-triangle hits. This supplies the finite obstruction-layer check supporting the regular exterior coherence mechanism.

E Bibliographic anchors

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